

Offense vs. Defense: Analyzing the Determinants of Team Success in Major League Soccer and Simulation of Probability of Winning the Champions of MLS cup Tournament

Introduction

As a soccer fan, I have long wondered which side of the game contributes more to winning: attacking quality or defensive solidity (including goalkeeping). In this project, I set out to answer that question using the rich statistical data available from Major League Soccer (MLS). I investigated the comparative impact of offensive versus defensive performance metrics on team success by combining the regular-season table with advanced team statistics from FBref (Major League Soccer section). Team success was quantified by outcomes such as points per game and final league position. Explanatory variables were grouped into two main blocks: (a) attacking metrics (e.g., goals scored, expected goals, shots, shots on target, and chance-creation measures) and (b) defensive and goalkeeping metrics (e.g., goals conceded, expected goals against, shots allowed, save percentage, and post-shot xG minus goals). I estimated separate regression models for the attack-only and defense-only blocks, as well as a combined model, and compared their performance using summary measures such as R^2 and RMSE.

To communicate these results, I used several visualizations to contrast offensive and defensive contributions to team success. Scatterplots of team success (e.g., points per game) versus key attacking and defensive metrics illustrated simple bivariate relationships. Coefficient plots with confidence intervals displayed and compared the estimated effects of offensive versus defensive variables within each regression model. In addition, bar charts comparing model performance (e.g., R^2 from offense-only vs defense-only) and residual-versus-fitted plots provided a concise summary of model fit and highlighted patterns that the models did not fully capture. Building on these descriptive and regression results, I then applied Poisson regression models to predict wins and losses using offensive and defensive/goalkeeping metrics with matches played as an offset, and fitted linear models to a logit-transformed winning percentage to compare the explanatory power of offense-only versus defense-only specifications. Finally, I used estimated team “strengths” from the Poisson win model to simulate the MLS Cup playoff bracket: using the actual Round One

matchups and a simple strength-based win-probability model, I simulated the tournament many times to obtain empirical championship probabilities for each qualified team. Together, these analyses provided a coherent framework to (a) compare the relative contributions of attacking and defensive performance to regular-season success and (b) translate those patterns into model-based estimates of each team's chance of winning the MLS Cup.

Method

Data and Variables

I used team-level data from one Major League Soccer (MLS) season, combining the regular-season table (wins, draws, losses, points) with advanced team statistics from FBref. Team success was quantified using points per game (PPG), total points, and winning percentage (WinPct). Explanatory variables were grouped into offensive metrics (e.g., goals, expected goals, shots, shots on target, chance-creation measures) and defensive/goalkeeping metrics (e.g., goals against, expected goals against, shots allowed, save percentage, and post-shot xG minus goals). All datasets were merged by team (Squad), and additional variables such as matches played (MP), PPG, and WinPct were derived from the W–D–L records.

Regression Analyses

I first conducted correlation analyses between W, D, L and each offensive, defensive, and possession-related variable to identify indicators most strongly associated with team performance. Then, I fit Poisson regression models to predict the number of wins and losses using offensive and defensive/goalkeeping metrics, respectively, with the log of matches played, log (MP), included as an offset to account for exposure. In parallel, I modeled team winning percentage using linear regression on a logit-transformed WinPct, estimating separate models for offense-only and defense-only predictor blocks. Model performance was compared using R^2 and descriptive residual diagnostics (e.g., residual-versus-fitted plots) to assess overall fit and potential patterns not captured by the models.

Simulation of MLS Cup Championship Probabilities

To link regular-season performance to postseason outcomes, I used the Poisson win model to derive an estimate of each team's "strength." For each team, I computed an expected number of wins from the Poisson model and converted this to an expected win rate

by dividing by matches played. In the playoff simulation, the actual Round One MLS Cup matchups were used as the starting bracket. For any given knockout match, the win probability for team A against team B was modeled as the ratio of its strength to the sum of both teams' strengths. Using these match-level probabilities, I simulated the entire playoff bracket many times (e.g., 10,000 iterations), recording the champion in each simulated tournament. The empirical proportion of simulations in which each team won the final was interpreted as that team's estimated probability of winning the MLS Cup.

Results

Poisson Regression

When modeling wins (W) with offensive variables in a Poisson regression (Figure 1), most attacking metrics were not individually significant once entered together, likely reflecting strong collinearity among shots, shots on target, and chance-creation measures. Two advanced efficiency indicators did emerge as significant predictors of wins: non-penalty xG per shot (off_npxG.Sh) and goals minus xG (off_G.xG) both had statistically significant coefficients ($p < .01$ and $p < .05$, respectively), indicating that teams that generated higher shot quality and outperformed their expected goals tended to achieve more wins, after controlling for overall shot volume.

In contrast, the Poisson model predicting losses (L) from defensive and goalkeeping variables did not yield any individually significant predictors (all $p > .05$, see Figure 2), even though the directions of coefficients were generally consistent with intuition (e.g., more goals against and shots on target against were associated with more losses, better save metrics with fewer losses). Finally, in the complementary linear models for logit-transformed winning percentage, the defense-only model explained slightly more variance than the offense-only model ($R^2 \approx 0.81$ vs. 0.79), suggesting that, at the team level, defensive/goalkeeping performance was at least as important as attacking performance for overall winning percentage in this MLS season.

Figure 1. The Results of Poisson Model of the Relationships Between Wins and Offensive Variables

```
Call:
glm(formula = formula_w_off, family = poisson(link = "log"),
     data = dat_w_off)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.08397    8.22299  -0.253  0.79993
off_sha      0.02566    0.78131   0.033  0.97380
off_sot      1.04806    2.10359   0.498  0.61833
off_sot      -0.02718    0.24394  -0.111  0.91127
off_g.sh     15.55524   19.96581   0.779  0.43592
off_g.sot     12.87977    7.65566   1.682  0.09249 .
off_fka      -0.18324    0.57945  -0.316  0.75183
off_npxg.sh -43.41361   16.64042  -2.609  0.00908 **
off_g.xg      -3.97701    1.57895  -2.519  0.01178 *
off_sca      -0.10187    0.26041  -0.391  0.69566
off_a.xag     1.21077    0.87654   1.381  0.16718
off_kp       -0.12398    0.26727  -0.464  0.64274
off_ppa      -0.01584    0.06572  -0.241  0.80958
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 54.992  on 29  degrees of freedom
Residual deviance: 13.009  on 17  degrees of freedom
AIC: 168.53

Number of Fisher Scoring iterations: 4
```

Figure 2. The Results of Poisson Model of the Relationships Between Loses and Defensive Variables

```
Call:
glm(formula = formula_L_def, family = poisson(link = "log"),
     data = dat_L_def)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   2.92179    16.46962   0.177   0.859
def_TkI       2.98878    3.36206   0.889   0.374
def_TkIw      -0.05486    0.25087  -0.219   0.827
def_Def.3rd   -2.91956    3.29257  -0.887   0.375
def_Mid.3rd   -2.83934    3.40663  -0.833   0.405
def_Att.3rd   -3.18941    3.46750  -0.920   0.358
def_Blocks    0.69974    3.14254   0.223   0.824
def_ShD       -0.56252    3.16612  -0.178   0.859
def_PassD     -0.63233    3.17966  -0.199   0.842
def_Int       0.09978    0.13021   0.766   0.443
def_Clr       -0.04366    0.04059  -1.076   0.282
def_Err       -0.19121    0.41709  -0.458   0.647
def_GA        -2.88756    3.59098  -0.804   0.421
def_SoTA      3.08089    2.03603   1.513   0.130
def_Saves     -2.88142    1.86247  -1.547   0.122
def_SavePer   -0.06519    0.14640  -0.445   0.656
def_CS        0.38651    2.48133   0.156   0.876
def_CK        -0.74796    2.03978  -0.367   0.714
def_PsXG      0.84850    3.36562   0.252   0.801
def_PsXG.sot -8.35390    18.31847  -0.456   0.648

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 35.3429  on 29  degrees of freedom
Residual deviance:  6.7605  on 10  degrees of freedom
AIC: 177.2

Number of Fisher scoring iterations: 4
```

Using the Poisson models, I compared the observed and predicted numbers of wins and losses for each team (see Figure 3 and 4). The scatterplots of observed vs. predicted W and L show that most clubs lie close to the 45-degree line, indicating that both the offensive-based win model and the defensive-based loss model reproduce the overall distribution of results reasonably well. A few teams deviate from the line, suggesting over- or under-performance relative to what would be expected based on their offensive or defensive statistics.

Figure 3. Observed vs. Poisson-predicted wins based on offensive metrics

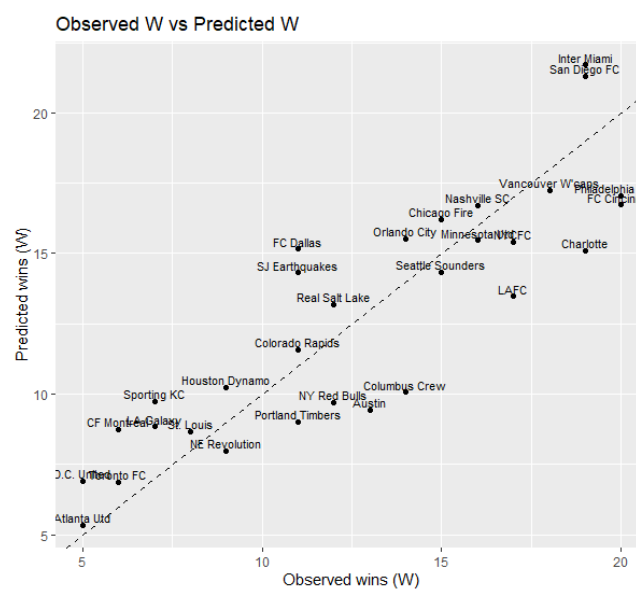
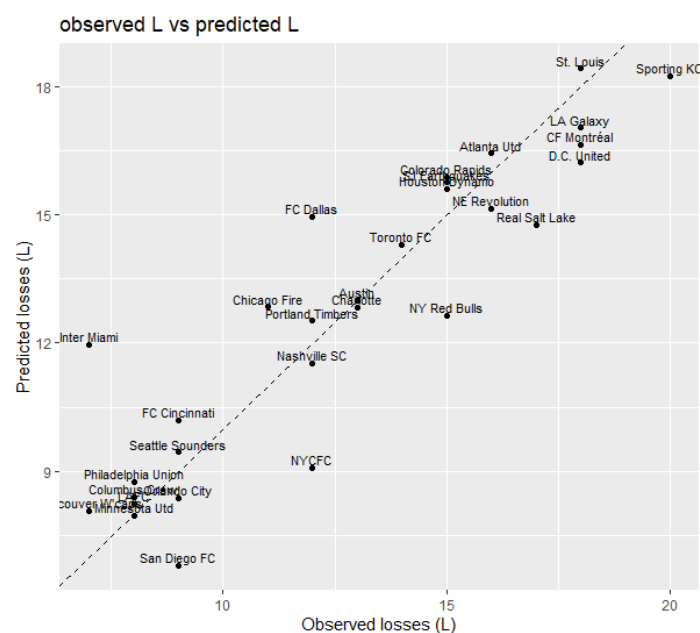


Figure 4. Observed vs. Poisson-predicted losses based on defensive and goalkeeping metrics



Using the offensive Poisson model, observed and predicted total points were strongly aligned overall, with most teams lying close to the 45-degree line (Figure 5). However, the residual plot (Figure 6) showed clear over- and under-performers relative to their offensive statistics: teams such as Columbus Crew, Charlotte, Austin, LAFC, and FC Cincinnati earned substantially more points than predicted (positive residuals), whereas FC Dallas, San Jose Earthquakes, Sporting KC, CF Montréal, and Inter Miami fell well below their model-based expectations (negative residuals).

Figure 5. Observed vs. predicted points from the offensive Poisson model

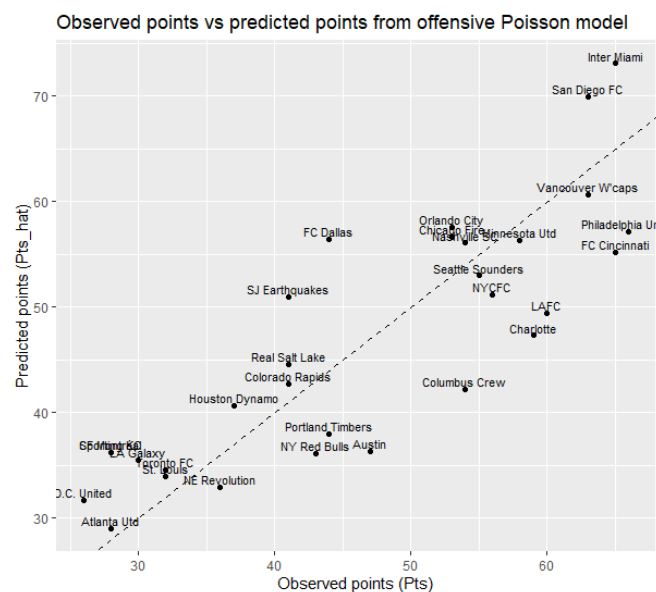
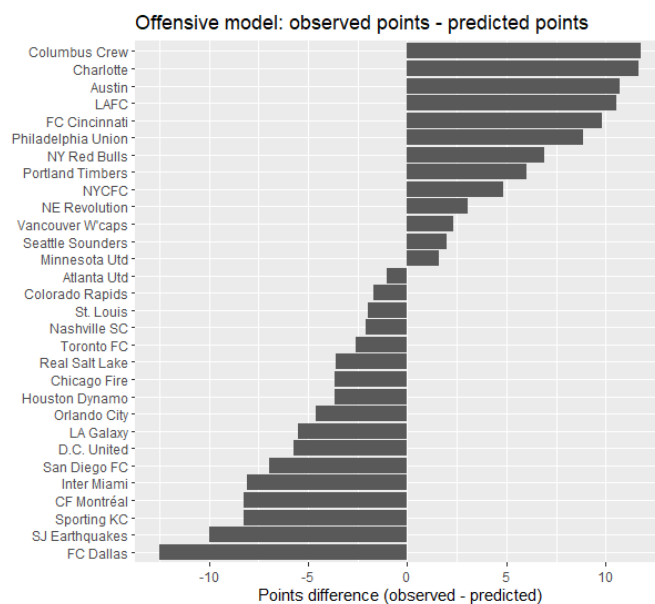


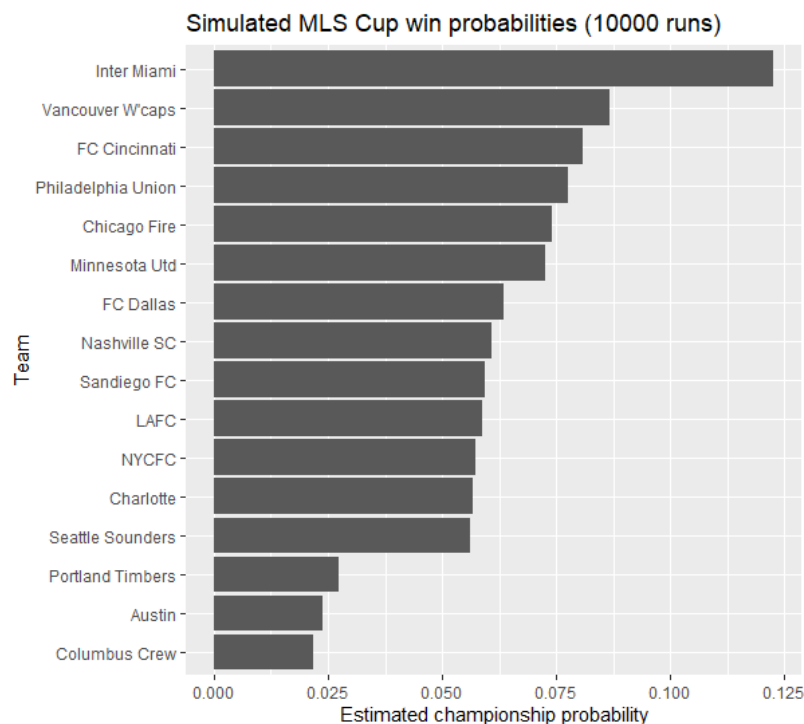
Figure 6. Points residuals from the offensive Poisson model (observed – predicted)



Simulations

Using the offensive Poisson win model as a measure of team strength, I simulated the MLS Cup playoff bracket 10,000 times with the actual Round One matchups. The resulting championship probabilities are summarized in Figure 7. Inter Miami emerged as the most likely champion, winning the simulated tournament in roughly 12% of runs. The next tier included Vancouver Whitecaps (about 9%) and FC Cincinnati (around 8%), followed by Philadelphia Union and Chicago Fire (each roughly 6–7%). Several other clubs—such as Minnesota United, FC Dallas, Nashville SC, San Diego FC, LAFC, and NYCFC—formed a middle group with probabilities in the 4–6% range. Teams like Portland Timbers, Austin, and Columbus Crew had the lowest estimated probabilities (roughly 1–3%). Overall, no team's simulated title chance exceeded about 12%, indicating a relatively wide and competitive field, even when using regular-season offensive performance to define team strength.

Figure 7. Simulated MLS Cup win probabilities based on 10,000 bracket simulations using Poisson-derived offensive strength.



Discussion

This project began from a simple fan question: is team success in Major League Soccer driven more by attacking quality or by defensive solidity (including goalkeeping)? Using one MLS season of team-level data, I compared offensive and defensive metrics as predictors of wins, losses, and winning percentage, and then translated these patterns into a simple simulation of MLS Cup championship probabilities.

The Poisson regressions provided a first look at how well each phase of play translated into results. When I modeled wins (W) as a Poisson outcome using offensive variables, most basic volume metrics (such as total shots and shots on target) were not significant once they were entered together, likely because these variables were highly collinear. Instead, more refined efficiency indicators—non-penalty xG per shot and goals minus xG—emerged as significant predictors. This pattern suggested that simply generating a high number of shots was not enough; what really separated teams in the win column was their ability to create high-quality chances and finish above expectation.

In contrast, the Poisson model predicting losses (L) from defensive and goalkeeping metrics did not produce any individually significant predictors, even though the estimated coefficients generally pointed in the expected directions (e.g., more goals against and shots on target against were associated with more losses, while better save metrics were associated with fewer losses). This lack of statistical significance likely reflected the small sample size at the team level and strong overlap among defensive indicators, which made it difficult to isolate the unique contribution of any single defensive variable.

To complement the Poisson models, I fit linear regressions of logit-transformed winning percentage using offense-only and defense-only predictor blocks. Here, the defense-only model explained slightly more variance in WinPct than the offense-only model ($R^2 \approx 0.81$ vs. 0.79), suggesting that defensive and goalkeeping performance were at least as important as attacking performance for sustaining a high winning percentage over the season. At the same time, observed-versus-predicted plots for wins, losses, and total points showed that both models captured the overall distribution of team performance reasonably well, while still leaving clear over- and under-performers. For example, some teams (such as Columbus and Charlotte) earned substantially more points than predicted from their offensive statistics, whereas others (such as FC Dallas and San Jose) fell well below their model-based

expectations. These discrepancies likely reflect factors not included in the models, such as schedule strength, injuries, match management, or randomness in close games.

Finally, I used the offensive Poisson win model to define a simple team “strength” and simulate the MLS Cup playoff bracket 10,000 times. The simulation produced a relatively flat distribution of championship probabilities: Inter Miami had the highest estimated chance at roughly 12%, with Vancouver, Cincinnati, Philadelphia, and Chicago forming the next tier, and no team emerging as a dominant favorite. This exercise illustrated how regular-season offensive performance can be translated into intuitive playoff odds, while also highlighting the inherent randomness of a knockout tournament—many teams retained a non-trivial chance to win the MLS Cup even after accounting for offensive strength. Notably, in reality Inter Miami did win the MLS Cup and faced Vancouver Whitecaps in the final. The fact that these two clubs also had the highest simulated championship probabilities in my Poisson-based bracket simulations suggests that the model produced reasonably credible forecasts, despite its relatively simple structure.

Several limitations should be acknowledged. The analysis relied on a single MLS season and a small number of teams, which limited statistical power and magnified the impact of multicollinearity among predictors. All models were based on aggregate team-level data, ignoring within-team variability, home-field advantage, schedule imbalance, and player-level information. In addition, the playoff simulation used offense-based strength only and assumed independence between matches, which simplifies real-world dynamics. Future work could extend this project by pooling multiple seasons, incorporating more detailed expected-goals match models, adjusting for strength of schedule, or using hierarchical models that simultaneously account for offensive and defensive strength at the team and player levels.